Introduction

Vũ Tuyết Trinh
SoICT - HUST

1

Outline

➢Introduction
- Examples of data integration applications
- Schema heterogeneity
- Goal of data integration
- Data integration architectures
- Review of basic database concepts

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG
School of Information and Communication Technology

2

2

Data Integration

- Databases are great: they let us manage huge amounts of data
  - Assuming you've put it all into your schema.
- In reality, data sets are often created independently
  - Only to discover later that they need to combine their data!
  - At that point, they're using different systems, different schemata and have limited interfaces to their data.
- The goal of data integration: tie together different sources, controlled by many people, under a common schema.

DBMS: it's all about abstraction

- *Logical* vs. *Physical*; *What* vs. *How.*

Students:

SSN	Name	Category
123-45-6789	Charles	undergrad
234-56-7890	Dan	grad
	…	…

Takes:

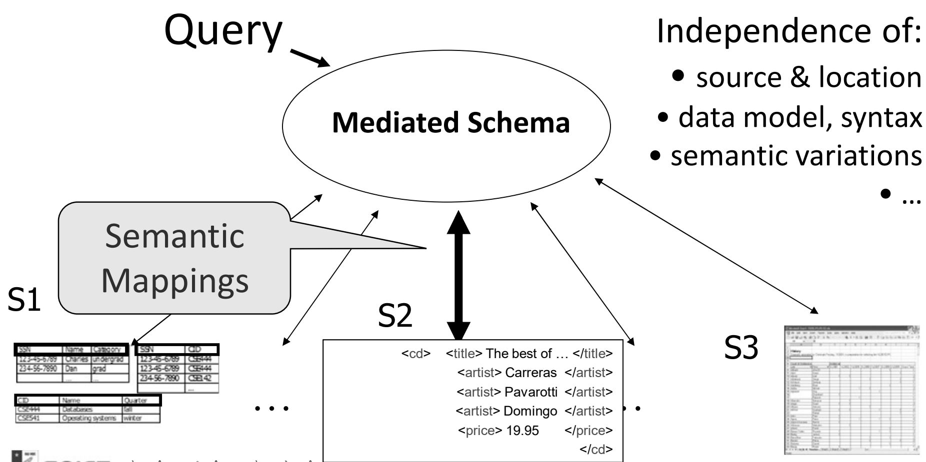
SSN	CID
123-45-6789	CSE444
123-45-6789	CSE444
234-56-7890	CSE142
	…

Courses:

CID	Name	Quarter
CSE444	Databases	fall
CSE541	Operating systems	winter

```
SELECT  C.name
FROM Students S, Takes T, Courses C
WHERE S.name="Mary" and
        S.ssn = T.ssn and T.cid = C.cid
```

Data Integration: A Higher-level Abstraction



Outline

✓Introduction: data integration as a new abstraction

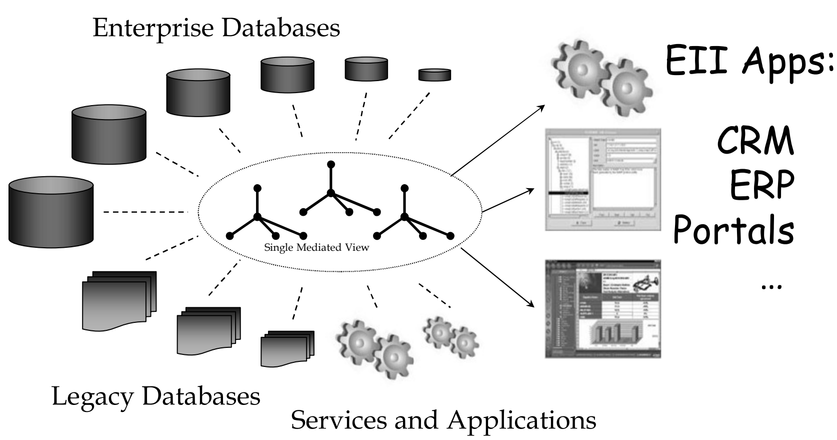
➢Examples of data integration applications

- Schema heterogeneity
- Goal of data integration, why it's a hard problem
- Data integration architectures
- Review of basic database concepts

Applications of Data Integration

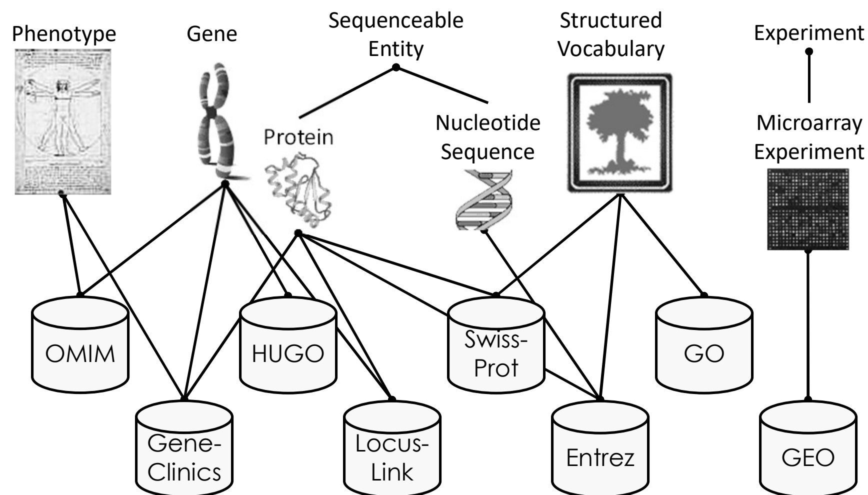
- Business
- Science
- Government
- The Web
- Pretty much everywhere

Application Area 1: Business



50% of all IT $$$ spent here!

Application Area 2: Science



Hundreds of biomedical data sources available; growing rapidly!

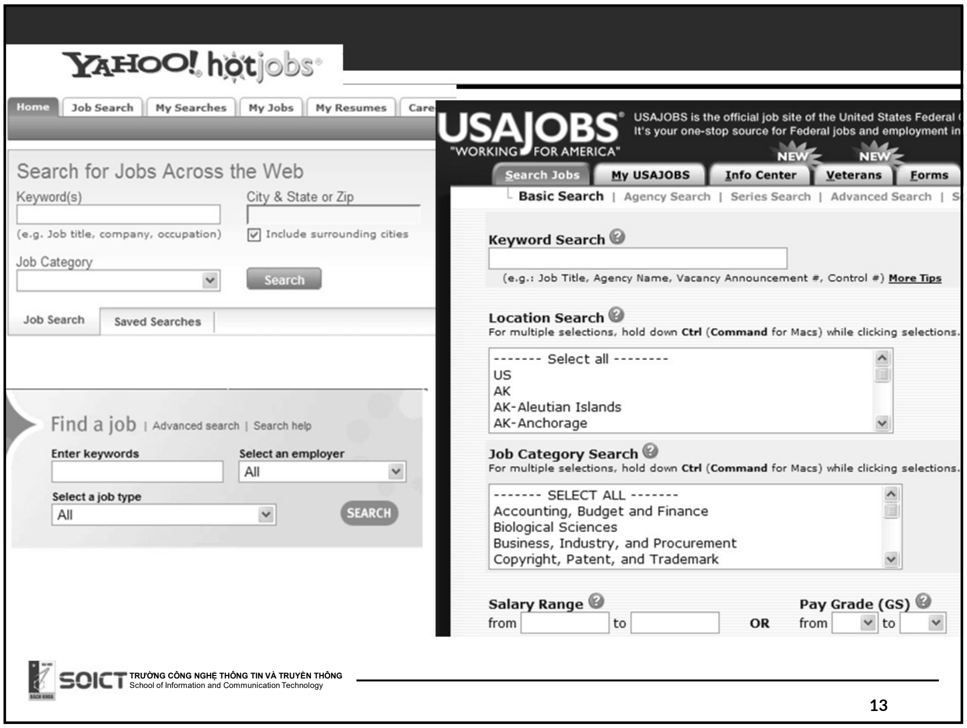
Application Area 3: The Web

11

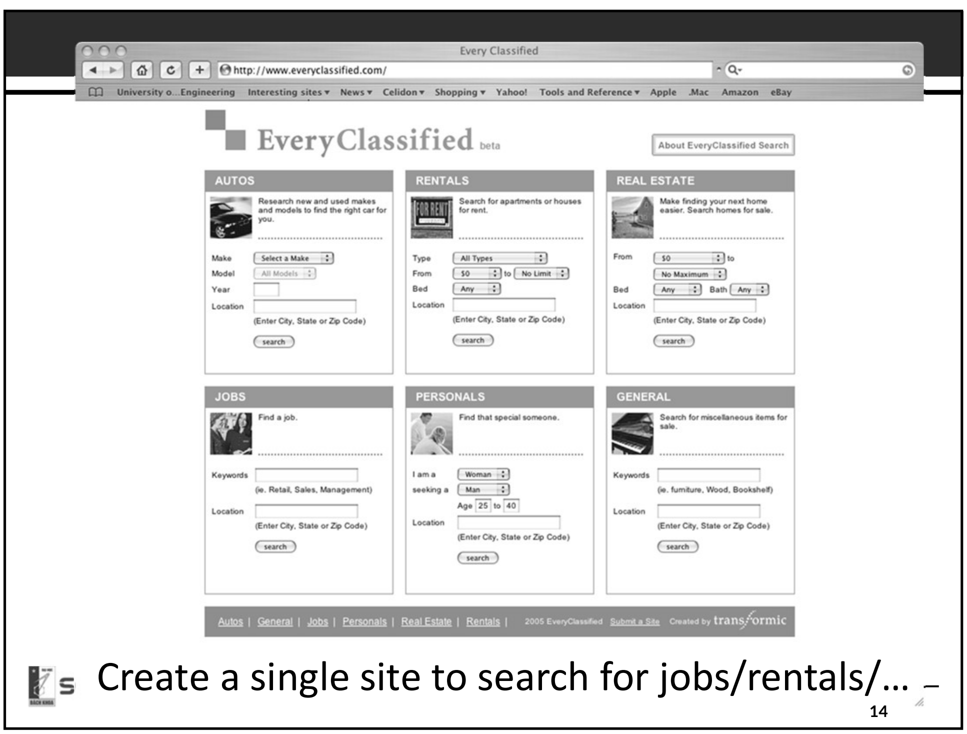
Web

- Millions of high quality HTML forms out there
- Each form has its own special interface
  - Hard to explore data across sites.
- Goal (for some domains):
  - A single interface into a multitude of web sources.

12

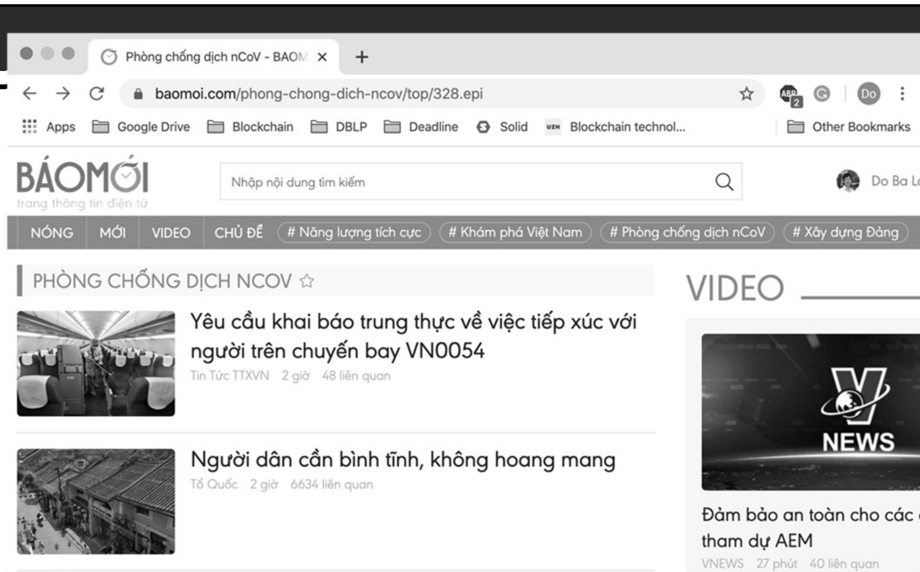

YAHOO! hotjobs
Home
Job Search
My Searches
My Jobs
My Resumes
Search for Jobs Across the Web
Keyword(s)
(e.g. Job title, company, occupation)
City & State or Zip
Include surrounding cities
Job Category
Search
Job Search
Saved Searches
Find a job
Advanced search
Search help
Enter keywords
Select an employer
All
Select a job type
All
SEARCH
USAJOBS
"WORKING FOR AMERICA"
USAJOBS is the official job site of the United States Federal
It's your one-stop source for Federal jobs and employment in
NEW
NEW
Search Jobs
My USAJOBS
Info Center
Veterans
Forms
Basic Search | Agency Search | Series Search | Advanced Search
Keyword Search
(e.g.: Job Title, Agency Name, Vacancy Announcement #, Control #) More Tips
Location Search
For multiple selections, hold down Ctrl (Command for Macs) while clicking selections.
------- Select all --------
US
AK
AK-Aleutian Islands
AK-Anchorage
Job Category Search
For multiple selections, hold down Ctrl (Command for Macs) while clicking selections.
------- SELECT ALL -------
Accounting, Budget and Finance
Biological Sciences
Business, Industry, and Procurement
Copyright, Patent, and Trademark
Salary Range
from
to
OR
Pay Grade (GS)
from
to
SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG
School of Information and Communication Technology
13

13


Every Classified
http://www.everyclassified.com/
EveryClassified beta
About EveryClassified Search
AUTOS
Research new and used makes and models to find the right car for you.
RENTALS
Search for apartments or houses for rent.
REAL ESTATE
Make finding your next home easier. Search homes for sale.
JOBS
Find a job.
PERSONALS
Find that special someone.
GENERAL
Search for miscellaneous items for sale.
Autos | General | Jobs | Personals | Real Estate | Rentals
Create a single site to search for jobs/rentals/…
14

14

Easily traverse between the site by clicking its name



Baomoi.com

Outline

- ✓Introduction: data integration as a new abstraction
- ✓Examples of data integration applications
- ➢Schema heterogeneity
- Goal of data integration, why it's a hard problem
- Data integration architectures
- Review of basic database concepts

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

17

17

Enterprise Data Integration: *FullServe Corporation*

Employees
- FullTimeEmp
- Hire
- TempEmployees

Training
- Courses
- Enrollments

Sales
- Products
- Sales

Resumes
- Interview
- CV

Services
- Services
- Customers
- Contracts

HelpLine
- Calls

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

18

18

EuroCard Corporation

Employees
- Employees
- Hire

Resumes
- Interview

Credit Cards
- Customer
- CustDetail

HelpLine
- Calls

Examples of Heterogeneity

FullServe

FullTimeEmp
ssn, empId, firstName
middleName, lastName

Hire
empId, hireDate, recruiter

TempEmployees
ssn, hireStart, hireEnd

EuroCard

Employees
ID, firstNameMiddleInitial,
lastName

Hire
ID, hireDate, recruiter

Find all employees (making over $100K)

Customer Call Center

Agents should have a full view of customer when they call in.

Sales

Products
Sales

Services

Services
Customers
Contracts

Credit Cards

Customer
CustDetail

Other Reasons to Integrate Data

- Create a (useful) web site for tracking services
- Collaborate with third parties
  - E.g., create branded services
- Comply with government regulations
  - Find "risky" employees
- Business intelligence
  - What's really wrong with our products?

Outline

- ✓Introduction: data integration as a new abstraction
- ✓Examples of data integration applications
- ✓Schema heterogeneity
- ➢Goal of data integration, why it's a hard problem
- Data integration architectures
- Review of basic database concepts

Goal of Data Integration

- Uniform query access to a set of data sources
- Handle:
  - Scale of sources: from tens to millions
  - Heterogeneity
  - Semi-structure

Why is it Hard?

- Systems-level reasons:
  - Managing different platforms
  - SQL across multiple systems is not so simple
  - Distributed query processing
- Logical reasons:
  - Schema (and data) heterogeneity
- 'Social' reasons:
  - Locating and capturing relevant data in the enterprise.
  - Convincing people to share (data fiefdoms)
    - Security, privacy and performance implications.

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

25

25

Data Integration Smorgasbord

Something for everyone:
- **Theory** of modeling data sources
- **Systems** aspects of data integration
- **Architectural** issues: e.g., P2P data sharing
- **AI** @ work: automated schema matching
- **Web**: latest on data integration & web
- **Commercial** products: BEA, IBM
- **Semantic Web**: what does it have to offer?
- New trends in DBMS: **uncertainty, dataspaces**

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

26

26

Outline

✓Introduction: data integration as a new abstraction

✓Examples of data integration applications

✓Schema heterogeneity

✓Goal of data integration, why it's a hard problem

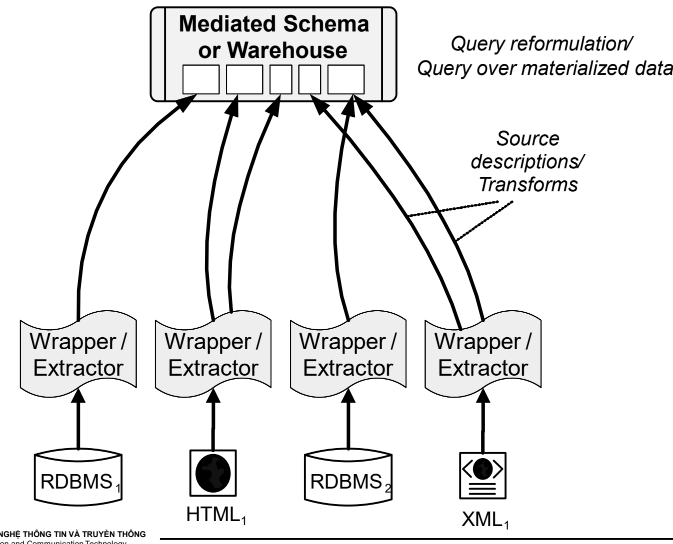
➢Data integration architectures

- Review of basic database concepts

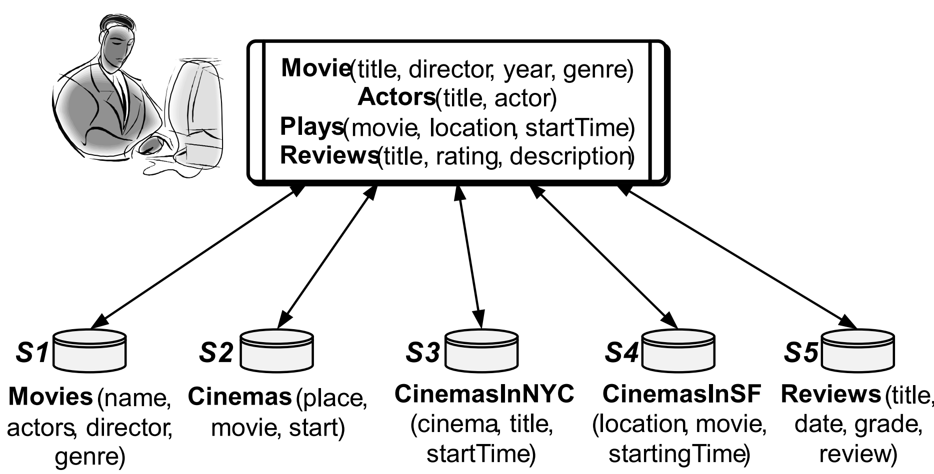
Virtual, Warehousing and in Between

- Data warehousing: integrate by bringing the data into a single physical warehouse
- Virtual data integration: leave the data at the sources and access it at query time.
- Some differences, but semantic heterogeneity arises in both cases.
- Numerous intermediate architectures.
- The course illustrates data integration technology mostly through the virtual architecture.

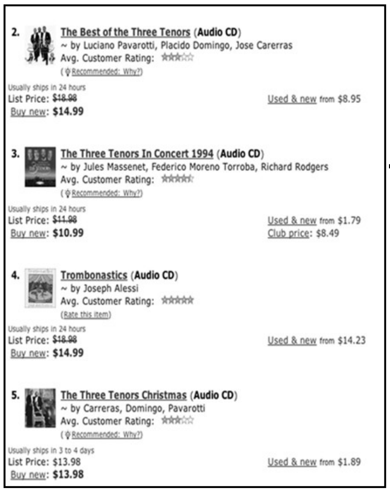
Virtual Data Integration Architecture



Example



Wrappers

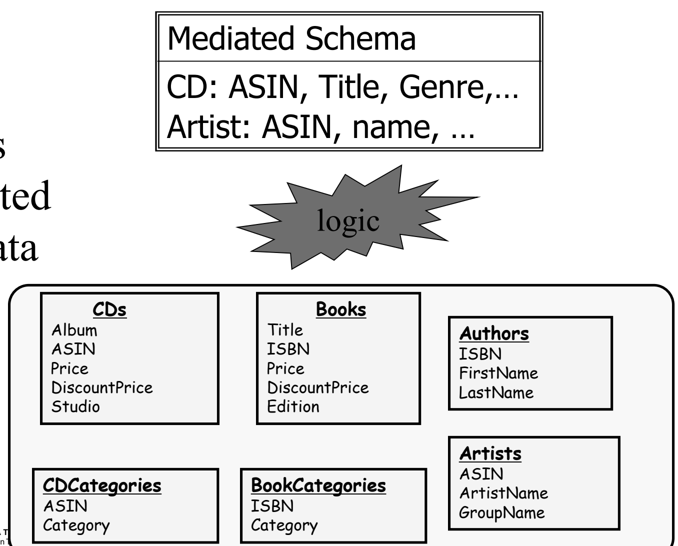


<cd> <title> The best of … </title>
<artist> Abiteboul </artist>
<artist> Pavarotti </artist>
<artist> Domingo </artist>
<price> 19.95 </price>
</cd>
…

Send queries to data sources and transform answers into tuples (or other internal data model).

Mediation Languages

Describe relationships between mediated schema and data sources



Woody Allen Comedies in NY

Mediated schema:

Movie: Title, director, year, genre
Actors: title, actor
Plays: movie, location, startTime
Reviews: title, rating, description

select **title, startTime**
from **Movie, Plays**
where Movie.title=Plays.movie AND
location="New York" AND
director="Woody Allen"

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

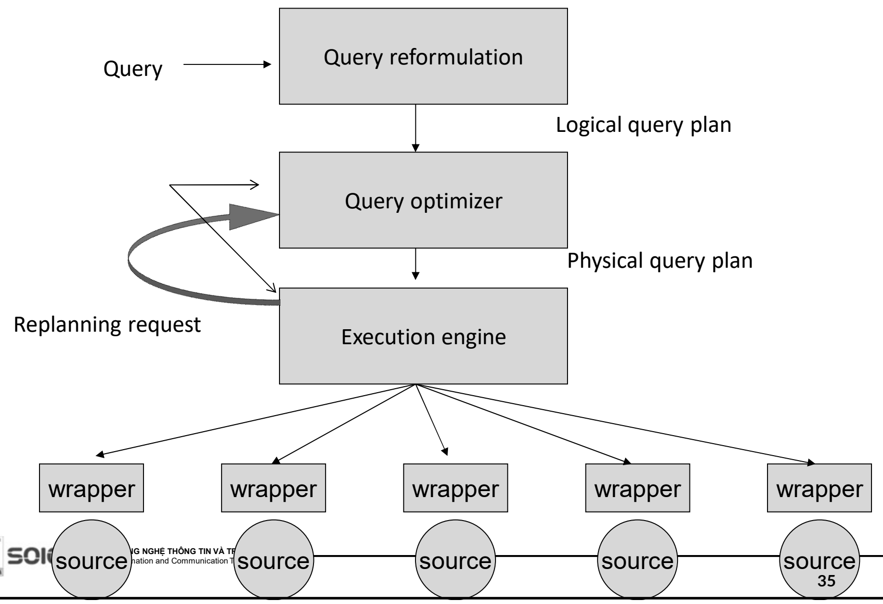
Movie: Title, director, year, genre
Actors: title, actor
Plays: movie, location, startTime
Reviews: title, rating, description

select **title, startTime**
from **Movie, Plays**
where Movie.title=Plays.movie AND
location="New York" AND
director="Woody Allen"

Sources S1 and S3 are relevant, sources S4 and S5 are irrelevant, and source S2 is relevant but possibly redundant.

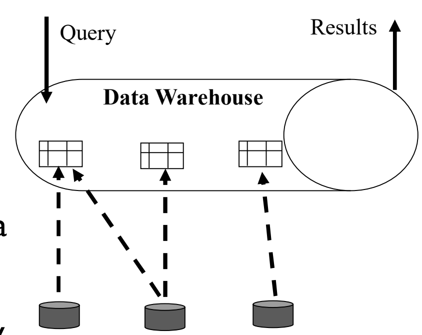
S1	S2	S3	S4	S5
Movies: name, actors, director, genre	Cinemas: place, movie, start	Cinemas in NYC: cinema, title, startTime	Cinemas in SF: location, movie, startingTime	Reviews: title, date grade, review

Query Processing



Data Warehouses – Offline Replication

- Determine physical schema
- Define a database with this schema
- Define procedural *mappings* in an "ETL tool" to import the data and clean it.
- Periodically copy all of the data from the data sources
  - Note that the sources and the warehouse are basically independent at this point



Pros and Cons of Data Warehouses

- ✗ Need to spend time to design the physical database layout, as well as logical
  - ✗ This actually takes a lot of effort!
- ✗ Data is generally not up-to-date (lazy or offline refresh)

- ✓ Queries over the warehouse don't disrupt the data sources
- ✓ Can run very heavy-duty computations, including data mining and cleaning

Outline

- ✓ Introduction: data integration as a new abstraction
- ✓ Examples of data integration applications
- ✓ Schema heterogeneity
- ✓ Goal of data integration, why it's a hard problem
- ✓ Data integration architectures
- ➢ Review of basic database concepts

Basic Database Concepts

- Relational data model
- Integrity constraints
- Queries and answers
- Conjunctive queries
- Datalog

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

39

39

Relational Terminology

Relational schemas
- Tables, attributes

Relation instances
- Sets (or multi-sets) of tuples

Integrity constraints
- Keys, foreign keys, inclusion dependencies

SOICT TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG School of Information and Communication Technology

40

40

Table/relation name

Attribute names

Product

PName	Price	Category	Manufacturer
Gizmo	$19.99	Gadgets	GizmoWorks
Powergizmo	$29.99	Gadgets	GizmoWorks
SingleTouch	$149.99	Photography	Canon
MultiTouch	$203.99	Household	Hitachi

Tuples or rows

SQL (very basic)

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

select recruiter, candidate
from **Interview**, **EmployeePerf**
where recuiter=name AND
EmployeePerf.grade < 2.5

Query Answers

- *Q(D):* the set (or multi-set) of rows resulting from applying the query *Q* on the database *D*.
- Unless otherwise stated, we will consider sets rather than multi-sets.

SQL (w/aggregation)

EmployeePerf:

empID, name, reviewQuarter, grade, reviewer

select reviewer, Avg(grade)
from **EmployeePerf**
where reviewQuarter="1/2007"

Notations

- A database instance: D
- Attributes: A, B, C.
- Lists of attributes: e.g., $\overline{A}$
- Relation names: R, S, and T
- Sets or lists of relations: $\overline{R}$
- Tuples with the lowercase letters: s, t.
- Restriction of the tuple t to the attributes in $\overline{A}$: $t^{\overline{A}}$

Integrity Constraints (Keys)

- A key is a set of columns that uniquely determine a row in the database:
  - There do not exist two tuples, t_1 and t_2 such that $t_1 \neq t_2$ and t_1 and t_2 have the same values for the key columns.
  - (EmpID, reviewQuarter) is a key for **EmployeePerf**

Integrity Constraints (Functional Dependencies)

- A set of attribute **A** functionally determines a set of attributes **B** if: whenever , t_1 and t_2 agree on the values of **A** , they must also agree on the values of **B**.

$$\overline{t_1^A} = \overline{t_2^A} \text{ then } \overline{t_1^B} = \overline{t_2^B}$$

- For example, (EmpID, reviewQuarter) functionally determine (grade).
- Note: a key dependency is a functional dependency where the key determines all the other columns.

Integrity Constraints (Foreign Keys)

- Given table **T** with key *B* and table **S** with key *A*: *A* is a foreign key of *B* in **T** if whenever a **S** has a row where the value of *A* is v, then **T** must have a row where the value of *B* is v.
- Example: the empID attribute of **EmployeePerf** is a foreign key for attribute emp of **Employee**.

General Integrity Constraints

Tuple generating dependencies (TGD's)

$$(\forall \bar{X}) s_1(\bar{X}_1), \ldots, s_m(\bar{X}_m) \rightarrow (\exists \bar{Y}) t_1(\bar{Y}_1), \ldots, t_l(\bar{Y}_l)$$

Equality generating dependencies (EGD's): right hand side contains only equalities.

$$(\forall \bar{Y}) t_1(\bar{Y}_1), \ldots, t_l(\bar{Y}_l) \rightarrow (Y_i^1 = Y_j^1), \ldots, (Y_i^k = Y_j^K)$$

General Integrity Constraints

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

Employee

empID	name	hireDate	manager
2335	Amanda Lucky	9/13/2005	Karina Lillberg
5443	Theodore Lanky	11/26/2004	Kasper Lillholm

$$(\forall I, N, R, Gr, Re)\ \text{EmployeePerformance}(I, N, R, Gr, Re) \rightarrow (\exists Na, Hd, Mg)\ \text{Employee}(I, Na, Hd, Ma)$$

General Integrity Constraints

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

$$(\forall C_1, D_1, R_1, H_1, G_1, D_2, R_2, H_2, G_2)$$
$$\text{Interview}(C_1, D_1, R_1, H_1, G_1), \text{Interview}(C_1, D_2, R_2, H_2, G_2) \rightarrow$$
$$D_1 = D_2, R_1 = R_2, H_1 = H_2, G_1 = G_2$$

General Integrity Constraints

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

$$(\forall I, N_1, R, G_1, Re_1, N_2, G_2, Re_2)$$
$$\text{EmployeePerformance}(I, N_1, R, G_1, Re_1), \text{EmployeePerformance}(I, N_2, R, G_2, Re_2)$$
$$\rightarrow G_1 = G_2$$

Conjunctive Queries

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

Q(Y,X) :-

Interview(X,D,Y,H,F), EmployeePerf(E,Y,T,W,Z),
W < 2.5.

select recruiter, candidate
from **Interview**, **EmployeePerf**
where recruiter=name AND
grade < 2.5

Unions of Conjunctive Queries

Multiple rules with the same head predicate express a union

Interview

candidate	date	recruiter	hireDecision	grade
Alan Jones	5/4/2006	Annette Young	No	2.8
Amanda Lucky	8/8/2008	Bob Young	Yes	3.7

EmployeePerformance

empID	name	reviewQuarter	grade	reviewer
2335	Amanda Lucky	1/2007	3.5	Eric Brown
5443	Theodore Lanky	2/2007	3.2	Bob Jones

Q(E,Y) :-

Interview(X,D,Y,H,F), EmployeePerf(E,Y,T,W,Z),
W < 2.5.

Q(E,Y) :-

Interview(X,D,Y,H,F), EmployeePerf(E,Y,T,W,Z),
Manager(y), W > 3.9.

Summary

- Data integration: abstract away the fact that data comes from multiple sources in varying schemata.
- Problem occurs everywhere: it's key to business, science, Web and government.
- Goal: reduce the effort involved in integrating.
- Regardless of the architecture, heterogeneity is a key issue.
- Architectures range from warehousing to virtual integration.